Previous Year Questions

Advanced (Functions)

Numerical

2025

Q1: Let $\mathbb{R}$ denote the set of all real numbers. Let $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to(0,4)$ be functions defined by $f(x)=\log_e\left(x^2+2x+4\right)$, and $g(x)=\dfrac{4}{1+e^{-2x}}$

Define the composite function fog^{-1} by $\left(fog^{-1}\right)(x)=f\left(g^{-1}(x)\right)$, where g^{-1} is the inverse of function g. Then the value of derivative of the composite function fog^{-1} at $x=2$ is _____.

Q2: Let $\mathbb{R}$ denote the set of all real numbers. Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x)>0$ for all $x\in\mathbb{R}$, and $f(x+y)=f(x)f(y)$ for all $x,y\in\mathbb{R}$. Let the real numbers $a_1,a_2,.....,a_{50}$ be in an arithmetic progression. If $f(a_{31})=64f(a_{25})$, and $\sum_{i=1}^{50}f(a_i)=3\left(2^{25}+1\right)$, then the value of $\sum_{i=6}^{30}f(a_i)$ is _____.

2024

Q1: Let $f:\mathbb{R}\to\mathbb{R}$ be a function such that $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{R}$ and $g:\mathbb{R}\to(0,\infty)$ be a function such that $g(x+y)=g(x)g(y)$ for all $x,y\in\mathbb{R}$. If $f\left(\dfrac{-3}{5}\right)=12$ and $g\left(\dfrac{-1}{3}\right)=2$, then the value of $\left(f\left(\dfrac{1}{4}\right)+g(-2)-8\right)g(0)$ is _____.

Q2: Let the function $f:\mathbb{R}\to\mathbb{R}$ be defined by

$f(x)=\frac{\sin x}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}+\frac{2}{e^{\pi x}}\frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)}$. Then the number of solutions of $f(x)=0$ in $\mathbb{R}$ is _____.

2020

Q1: Let the function $f:[0,1]\to\mathbb{R}$ be defined by $f(x)=\frac{4^x}{4^x+2}$. Then the value of

$f\left(\frac{1}{40}\right)+f\left(\frac{2}{40}\right)+f\left(\frac{3}{40}\right)+.....+f\left(\frac{39}{40}\right)-f\left(\frac{1}{2}\right)$ is _____.

Q2: Let the function $f:(0,\pi)\to\mathbb{R}$ be defined by $f(\theta)=(\sin\theta+\cos\theta)^2+(\sin\theta-\cos\theta)^4$. Suppose the function f has a local minimum at θ precisely when $\theta\in(\lambda_1\pi,.....,\lambda_r\pi)$, where $0<\lambda_1<...\lambda_r<1$. Then the value of $\lambda_1+...+\lambda_r$ is _____.

Q3: Let $f:[0,2]\to\mathbb{R}$ be a function defined by $f(x)=(3-\sin(2\pi x))\sin\left(\pi x-\frac{\pi}{4}\right)-\sin\left(3\pi x+\frac{\pi}{4}\right)$. If $\alpha,\beta\in[0,2]$ are such that $\{x\in[0,2]:\ f(x)\geq 0\}=[\alpha,\beta]$, then the value of $\beta-\alpha$ is _____.

Q4: For a polynomial g(x) with real coefficients, let m_g denote the number of distinct real roots of g(x). Suppose S is the set of polynomials with real coefficients defined by

$S=\left\{\left(x^2-1\right)^2\left(a_0+a_1x+a_2x^2+a_3x^3\right):a_0,a_1,a_2,a_3\in\mathbb{R}\right\}$; For a polynomial f, let f' and f'' denote its first and second order derivatives, respectively. Then the minimum possible value of ($m_{f'}$ + $m_{f''}$), where $f\in S$, is _____.

2018

Q1: Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X, then the value of $\frac{1}{5!}(\beta-\alpha)$ is _____.

2009

Q1: If the function $f(x)=x^3+e^{x/2}$ and $g(x)=f^{-1}(x)$, then the value of $g'(1)$ is _____.

MCQ (More than One Correct Answer)

2025

Q1: Let $\mathbb{N}$ denote the set of all natural numbers, and $\mathbb{Z}$ denote the set of all integers. Consider the functions $f: \mathbb{N} \to \mathbb{Z}$ and $g: \mathbb{Z} \to \mathbb{N}$ defined by

$$f(n) = \begin{cases} \dfrac{(n+1)}{2} & if \ldots n \ldots is \ldots odd \\ \dfrac{(4-n)}{2} & if \ldots n \ldots is \ldots even, \end{cases}$$

and

$$g(n) = \begin{cases} 3+2n & if \ldots n \geq 0, \\ -2n & if \ldots n < 0. \end{cases}$$

Define $(gof)(n) = g(f(n))$ for all $n \in \mathbb{N}$ and $(fog)(n) = f(g(n))$ for all $n \in \mathbb{Z}$.

Then which of the statements is (are) true?

(a) gof is NOT one-one and gof is NOT onto.
(b) fog is NOT one-one but fog is onto.
(c) g is one-one and g is onto.
(d) f is NOT one-one but f is onto.

2023

Q1: Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

(a) There are infinitely many functions from S to T.
(b) There are infinitely many strictly increasing functions from S to T.
(c) The number of continuous functions from S to T is is at most 120.
(d) Every continuous function from S to T is differentiable.

Q2: Let $f:[0,1] \to [0,1]$ be the function defined by $f(x) = \frac{x^3}{3} - x^2 + \frac{5}{9}x + \frac{17}{36}$. Consider the square region $S = [0,1] \times [0,1]$. Let $G = \{(x,y) \in S : y > f(x)\}$ be called the green region and

$R=\{(x,y)\in S : y< f(x)\}$ be called the red region. Let $L_h=\{(x,h)\in S : x\in[0,1]\}$ be the horizontal line drawn at a height $h\in[0,1]$. Then which of the statements is(are) true?

(a) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the green region below the line L_h.

(b) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the red region below the line L_h.

(c) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the green region above the line L_h equals the area of the red region below the line L_h.

(d) There exists an $h\in\left[\frac{1}{4},\frac{2}{3}\right]$ such that the area of the red region above the line L_h equals the area of the green region below the line L_h.

2022

Q1: Let $|M|$ denote the determinant of a square Matrix M. Let $g:\left[0,\frac{\pi}{2}\right]\to\mathbb{R}$ be the function defined by $g(\theta)=\sqrt{f(\theta)-1}+\sqrt{f\left(\frac{\pi}{2}-\theta\right)-1}$ where

$$f(\theta)=\frac{1}{2}\begin{bmatrix}1 & \sin\theta & 1\\ -\sin\theta & 1 & \sin\theta\\ -1 & -\sin\theta & 1\end{bmatrix}+\begin{vmatrix}\sin\pi & \cos\left(\theta+\frac{\pi}{4}\right) & \tan\left(\theta-\frac{\pi}{4}\right)\\ \sin\left(\theta-\frac{\pi}{4}\right) & -\cos\frac{\pi}{2} & \log_e\left(\frac{4}{\pi}\right)\\ \cot\left(\theta+\frac{\pi}{4}\right) & \log_e\left(\frac{\pi}{4}\right) & \tan\pi\end{vmatrix}$$

. Let $p(x)$ be a quadratic polynomial whose roots are the maximum and minimum values of the function $g(\theta)$, and $p(2)=2-\sqrt{2}$. Then which of the following is (are) true?

(a) $p\left(\frac{3+\sqrt{2}}{4}\right)<0$

(b) $p\left(\frac{1+3\sqrt{2}}{4}\right)>0$

(c) $p\left(\frac{5\sqrt{2}-1}{4}\right)>0$ (d) $p\left(\frac{5-\sqrt{2}}{4}\right)<0$

2015

Q1: Let $f(x)=\sin\left(\frac{\pi}{6}\sin\left(\frac{\pi}{2}\sin x\right)\right)$ for all $x\in\mathbb{R}$ and $g(x)=\frac{\pi}{2}\sin x$ for all $x\in\mathbb{R}$. Let $(fog)(x)$ denote $f(g(x))$ and $(gof)(x)$ denote $g(f(x))$. The which of the following is/are true?

(a) Range of f is $\left[-\frac{1}{2},\frac{1}{2}\right]$ (b) Range of fog is $\left[-\frac{1}{2},\frac{1}{2}\right]$

(c) $\lim_{x\to 0}\frac{f(x)}{g(x)}=\frac{\pi}{6}$ (d) There is an $x\in\mathbb{R}$ such that $(gof)(x)=1$.

2014

Q1: For every pair of continuous function $f,g:[0,1]\to\mathbb{R}$ such that $\max\{f(x):x\in[0,1]\}=\max\{g(x):x\in[0,1]\}$. The correct statements is(are)

(a) $[f(c)]^2+3f(c)=[g(c)]^2+3g(c)$ for some $c\in[0,1]$

(b) $[f(c)]^2+f(c)=[g(c)]^2+3g(c)$ for some $c\in[0,1]$

(c) $[f(c)]^2+3f(c)=[g(c)]^2+g(c)$ for some $c\in[0,1]$

(d) $[f(c)]^2=[g(c)]^2$ for some $c\in[0,1]$

Q2: Let $f:\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to\mathbb{R}$ be given by $f(x)=[\log(\sec x+\tan x)]^3$ .Then,

(a) $f(x)$ is an odd function (b) $f(x)$ is a one-one function

(c) $f(x)$ is an onto function (d) $f(x)$ is an even function

2012

Q1: Let $f:(-1,1)\to\mathbb{R}$ be such that $f(\cos 4\theta)=\dfrac{2}{2-\sec^2\theta}$ for $\theta\in\left(0,\dfrac{\pi}{4}\right)\cup\left(\dfrac{\pi}{4},\dfrac{\pi}{2}\right)$. Then the value(s) of $f\left(\dfrac{1}{3}\right)$ is(are):

(a) $1-\sqrt{\dfrac{3}{2}}$

(b) $1+\sqrt{\dfrac{3}{2}}$

(c) $1-\sqrt{\dfrac{2}{3}}$

(d) $1+\sqrt{\dfrac{2}{3}}$

2011

Q1: Let $f:(0,1)\to\mathbb{R}$ be defined by $f(x)=\dfrac{b-x}{1-bx}$, where b is a constant such that $0<b<1$. Then

(a) f is not invertible on $(0,1)$

(b) $f\neq f^{-1}$ on $(0,1)$ and $f'(b)=\dfrac{1}{f'(0)}$.

(c) $f=f^{-1}$ on $(0,1)$ and $f'(b)=\dfrac{1}{f'(0)}$.

(d) f^{-1} is differentiable on $(0,1)$.

MCQ (Single Correct Answer)

2020

Q1: If the function $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=|x|(x-\sin x)$, then which of the following statements is TRUE?

(a) f is one-one, but NOT onto.
(b) f is onto, but NOT one-one.
(c) f is BOTH one-one and onto.
(d) f is NEITHER one-one NOR onto.

2018

Q1: Let $E_1 = \left\{ x \in \mathbb{R} : x \neq 1 \ldots and \ldots \frac{x}{x-1} > 0 \right\}$ and $E_2 = \left\{ \begin{array}{l} x \in E_1 : \sin^{-1}\left(\log_e \left(\frac{x}{x-1} \right) \right) \\ is \ldots a \ldots real \ldots number \end{array} \right\}$.

(Here the inverse trigonometric function $\sin^{-1} x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.).Let $f : E_1 \to \mathbb{R}$ be the function defined by $f(x) = \log_e \left(\frac{x}{x-1} \right)$ and $g : E_2 \to \mathbb{R}$ be the function defined by $g(x) = \sin^{-1}\left(\log_e \left(\frac{x}{x-1} \right) \right)$.

List-I	List-II
P. The range of f is	1. $\left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$
Q. The range of g contains	2. $(0,1)$
R. The domain of f contains	3. $\left[-\frac{1}{2}, \frac{1}{2} \right]$
S. The domain of g is	4. $(-\infty, 0) \cup (0, \infty)$
	5. $\left(-\infty, \frac{e}{e-1} \right)$
	6. $(-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{e-1} \right]$

The correct option is:

(a) $P \to 4; Q \to 2; R \to 1; S \to 1$
(b) $P \to 3; Q \to 3; R \to 6; S \to 5$
(c) $P \to 4; Q \to 2; R \to 1; S \to 6$
(d) $P \to 4; Q \to 3; R \to 6; S \to 5$

2017

Q1: Let $S = \{1, 2, 3, \ldots, 9\}$. For $k = 1, 2, \ldots, 5$,let N_k be the number of subsets of S, each containing five elements out of which exactly k are odd. Then $N_1 + N_2 + N_3 + N_4 + N_5 =$

(a) 210 (b) 252

(c) 126 (d) 125

2014

Q1: Let $f_1 \to \mathbb{R} \to \mathbb{R}$, $f_2 : [0,\infty) \to \mathbb{R}$, $f_3 \to \mathbb{R} \to \mathbb{R}$ and $f_4 \to \mathbb{R} \to [0,\infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & if \ldots x < 0, \\ e^x & if \ldots x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & if \ldots x < 0, \\ x & if \ldots x \geq 0; \end{cases}$$

and

$$f_4(x) = \begin{cases} f_2(f_1(x)) & if \ldots x < 0, \\ f_2(f_1(x)) - 1 & if \ldots x \geq 0; \end{cases}$$

List-I	List-II
P. f_4 is	(i) onto but not one-one
Q. f_3 is	(ii) neither continuous nor one-one
R. $f_2 o f_1$ is	(iii) differentiable but not one-one
S. f_2 is	(iv) continuous and one-one

(a) $P \to 3, Q \to 1, R \to 4, S \to 2$
(b) $P \to 1, Q \to 3, R \to 4, S \to 2$
(c) $P \to 3, Q \to 1, R \to 2, S \to 4$
(d) $P \to 1, Q \to 3, R \to 2, S \to 4$

2012

Q1: The function $f : [0,3] \to [1,29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(a) one-one and onto.
(b) onto but not one-one.
(c) one-one but not onto.
(d) neither one-one nor onto.

2011

Q1: Let $f(x)=x^2$ and $g(x)=\sin x$ for all $x\in\mathbb{R}$. Then the set of all x satisfying $(fogogof)(x)=(gogof)(x)$, where $(fog)(x)=f(g(x))$, is

(a) $\pm\sqrt{n\pi}, n\in\{0,1,2,.....\}$

(b) $\pm\sqrt{n\pi}, n\in\{1,2,.....\}$

(c) $\frac{\pi}{2}+2n\pi, n\in\{....,-2,-1,0,1,2,....\}$

(d) $2n\pi, n\in\{.....,-2,-1,0,1,2,....\}$

Q2: Match the statements given in Column I with the intervals/union of intervals given in Column II:

Column I	Column II
(A) The set $\left\{\mathrm{Re}\left(\frac{2iz}{1-z^2}\right): z \text{ is a complex number}, \lvert z\rvert=1,\ z\neq\pm1\right\}$ is	(p) $(-\infty,-1)\cup(1,\infty)$
(B) The domain of the function $f(x)=\sin^{-1}\left(\frac{8(3)^{x-2}}{1-3^{2(x-1)}}\right)$ is	(q) $(-\infty,0)\cup(0,\infty)$
(C) If $f(\theta)=\begin{vmatrix}1 & \tan\theta & 1\\ -\tan\theta & 1 & \tan\theta\\ -1 & -\tan\theta & 1\end{vmatrix}$, then the set $\left\{f(\theta): 0\le\theta<\frac{\pi}{2}\right\}$ is	(r) $[2,\infty)$
(D) If $f(x)=x^{3/2}(3x-10)$, $x\ge0$, then $f(x)$ is increasing in	(s) $(-\infty,-1]\cup[1,\infty)$
	(t) $(-\infty,0]\cup[2,\infty)$

(a) $A\to S, B\to T, C\to P, D\to Q$

(b) $A\to S, B\to T, C\to R, D\to P$

(c) $A\to S, B\to T, C\to R, D\to R$

(d) $A\to P, B\to Q, C\to R, D\to R$

2010

Q1: Let f, g and h be real valued functions defined on the interval $[0,1]$ by

$f(x)=e^{x^2}+e^{-x^2}$

$g(x) = xe^{x^2} + e^{-x^2}$

and $h(x) = x^2 e^{x^2} + e^{-x^2}$.

If a, b and c denote, respectively, the absolute maximum of f, g and h on $[0,1]$, then

(a) $a = b$ and $c \neq b$

(b) $a = c$ and $a \neq b$

(c) $a \neq b$ and $c \neq b$

(d) $a = b = c$

Q2: Let $S = \{1, 2, 3, 4\}$. The total number of unordered pairs of disjoint subsets of S is equal to :

(a) 25

(b) 34

(c) 42

(d) 41

2006

Q1: If $f''(x) = -f(x)$ and $g(x) = f'(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given that $F(5) = 5$, then $F(10)$ is equal to:

(a) 5

(b) 10

(c) 0

(d) 15

2005

Q1: Find the range of value of t for which $2\sin t = \frac{1 - 2x + 5x^2}{3x^2 - 2x - 1}, t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(a) $\left[-\frac{\pi}{3}, -\frac{\pi}{10}\right] \cup \left[\frac{\pi}{10}, \frac{\pi}{2}\right]$

(b) $\left[-\frac{\pi}{2}, -\frac{\pi}{10}\right] \cup \left[\frac{3\pi}{10}, \frac{\pi}{2}\right]$

(c) $\left[-\frac{\pi}{2}, -\frac{\pi}{6}\right] \cup \left[\frac{3\pi}{10}, \frac{\pi}{3}\right]$

(d) $\left[\frac{\pi}{2}, -\frac{\pi}{10}\right] \cup \left[\frac{\pi}{10}, \frac{\pi}{2}\right]$